



DC – AI – Routing – Switching – Security – Wireless – Automation

BIER Overlay

An HPE Juniper Networking Techpost

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HPE



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1 BIER Overlay

Suneesh Babu - 08/08/2024

The series is composed of the following posts:

- [Introduction to BIER and BIER Underlay](#)
- [BIER Table Lookup](#)
- BIER Overlay (present article)

In this TechPost, we talk in deeper detail about overlay details.

1.1 BIER overlay

The Overlay helps determine interested multicast sources and listeners in the BIER domain. The Overlay can be MVPN, EVPN, or Global IP Multicast. BIER on PTX10002-36QDD supports MVPN as the Overlay in Junos 23.4R2 release.

MVPN can use P2MP tunnels created by RSVP-TE, mLDP, or PIM to transport customer multicast traffic across the provider's backbone network. P2MP tunnels in "aggregate tunnel mode" allocate an "upstream-assigned MPLS label" for each VPN and each packet sent on this P2MP tunnel carries the upstream-assigned MPLS label that the ingress PE has bound to the packet's VPN.

The upstream-assigned label procedures can result in scaling problems, as each PE allocates a label from its own label space. A detailed analysis is captured in the [mvpn-evpn-aggregation-label draft](#). BIER implementation on PTX10002-36QDD follows the optimized approach mentioned in the above draft to solve all scaling issues. Here, the labels are allocated in the provider network by a central entity. All PEs use the same label to represent a VPN.

The BIER sub-domain got mapped to MVPN VPN instance as shown in the output below:

```
1 regress@r1-sb-bier-RE0> show configuration routing-instances m_customer_A |display inheritance
2 no-comments |no-more
3 instance-type vrf;
4 protocols {
5     bgp {
6         group ce-pe-v4-customer-1 {
7             type external;
8             family inet {
9                 unicast;
10            }
11            export Export-Local-Routes;
12            peer-as 64001;
13            local-as 64512;
14            as-override;
15            neighbor 13.1.1.2;
16        }
17    }
18    mvpn {
19        mvpn-mode {
20            spt-only;
21        }
22    }
23    pim {
24        rp {
25            static {
26                address 100.1.1.1;
27                address 2100:100:1:1::1;
28            }
29        }
30        interface et-0/0/18.10;
31        interface lo0.1;
```

```

31 }
32 }
33 interface et-0/0/18.10;
34 interface lo0.1;
35 route-distinguisher 11.1.1.1:1;
36 vrf-target target:64512:1;
37 vrf-table-label static 990001;
38 provider-tunnel {
39     bier {
40         subdomain-id 100;
41         label 990001;
42     }
43 }
44
45 regress@r1-sb-bier-RE0>

```

1.1.1 BGP MVPN Peering on PEs

The BFIR/BFER routers are enabled with BGP MVPN family, and BFR R2 acts as the Route Reflector. The BGP Peering details:

```

1 regress@r2-sb-bier-RE0> show bgp summary
2
3 Warning: License key missing; requires 'BGP' license
4
5 Threading mode: BGP I/O
6 Default eBGP mode: advertise - accept, receive - accept
7 Groups: 1 Peers: 5 Down peers: 0
8 Table Tot Paths Act Paths Suppressed History Damp State Pending
9 bgp.mvpn.0
10          9 6 0 0 0 0
11 bgp.l3vpn.0
12          10 10 0 0 0 0
13 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
14 11.1.1.1 64512 2797 2781 0 0 20:58:57 Establ
15   bgp.mvpn.0: 1/1/1/0
16   bgp.l3vpn.0: 2/2/2/0
17 11.1.1.4 64512 2797 2777 0 1 20:58:41 Establ
18   bgp.mvpn.0: 2/2/2/0
19   bgp.l3vpn.0: 2/2/2/0
20 11.1.1.5 64512 2808 2780 0 1 20:58:25 Establ
21   bgp.mvpn.0: 1/2/2/0
22   bgp.l3vpn.0: 2/2/2/0
23 11.1.1.6 64512 2807 2777 0 1 20:58:06 Establ
24   bgp.mvpn.0: 1/2/2/0
25   bgp.l3vpn.0: 2/2/2/0
26 11.1.1.7 64512 2806 2775 0 1 20:57:46 Establ
27   bgp.mvpn.0: 1/2/2/0
28   bgp.l3vpn.0: 2/2/2/0
29
30 regress@r2-sb-bier-RE0>

```

1.1.2 Multicast Source and Receivers Joining the PE's CE

The multicast source is emulated on the IXIA behind R1 and listeners are emulated behind R4, R5, R6 and R7.

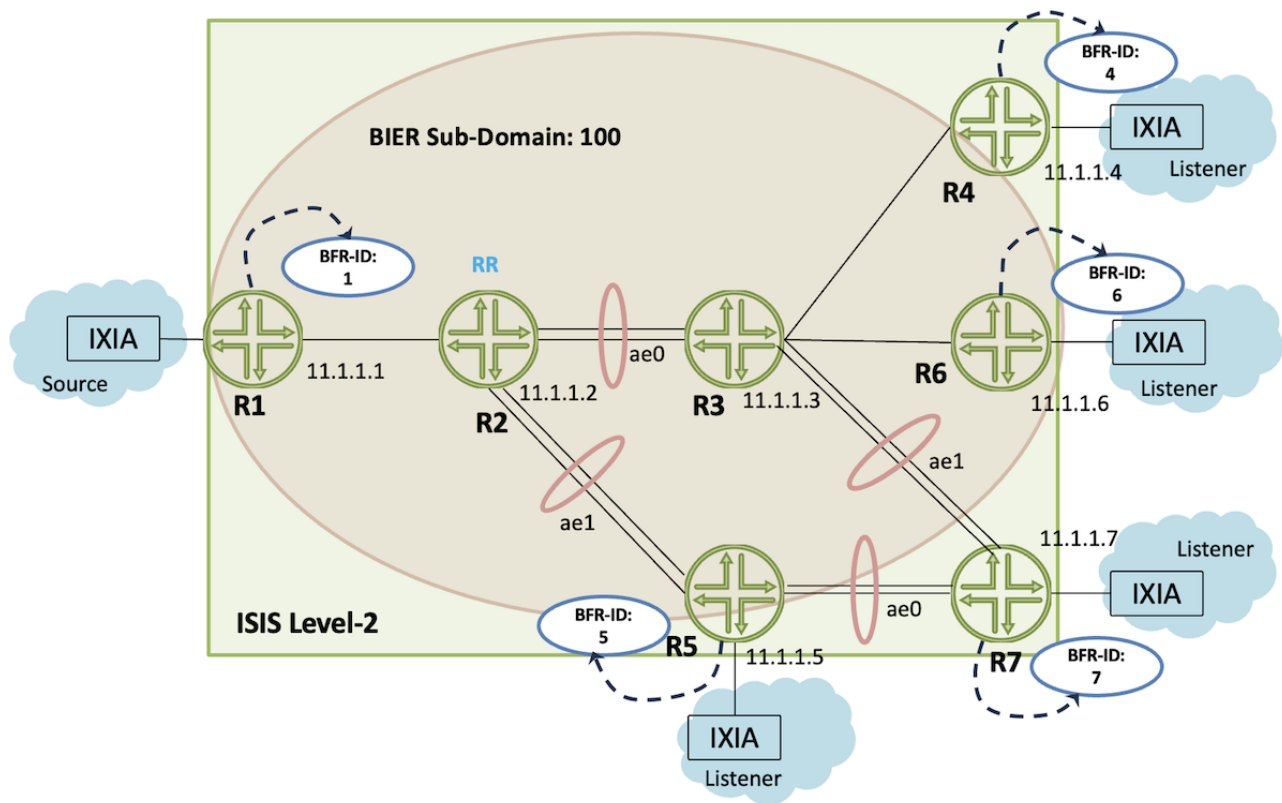


Figure 1: Multicast Source and Listeners Joining the Topology

Source 13.1.1.2 is residing at CE1 of R1 and Listeners are at R4, R5, R6, and R7. The listeners send igmp join for (13.1.1.2, 232.1.1.1).

1.1.3 Remote Receiver at R4

At BFER R4, igmp is enabled. Let's check the group membership.

```

1 regress@r4-sb-bier-RE0> show igmp group
2
3 Warning: License key missing; requires 'L3 Multicast/IGMP' license
4
5 Interface: et-0/0/18.10, Groups: 1
6   Group: 232.1.1.1
7     Group mode: Include
8     Source: 13.1.1.2
9     Last reported by: 13.1.2.2
10    Timeout: 240 Type: Dynamic
11 Interface: local, Groups: 4
12   Group: 224.0.0.0
13     Source: 0.0.0.0
14     Last reported by: Local
15     Timeout: 0 Type: Dynamic
16   Group: 224.0.0.2
17     Source: 0.0.0.0
18     Last reported by: Local
19     Timeout: 0 Type: Dynamic
20   Group: 224.0.0.13
21     Source: 0.0.0.0
22     Last reported by: Local
23     Timeout: 0 Type: Dynamic
24   Group: 224.0.0.22
25     Source: 0.0.0.0
26     Last reported by: Local
27     Timeout: 0 Type: Dynamic
28
29 regress@r4-sb-bier-RE0>

```

1.1.4 Multicast Source at R1

R1 acts as the BFIR router where the multicast source is present. As shown below, the down-stream OIF is the Pseudo-MVPN interface corresponding to the BIER tunnel and IIF is the directly connected interface from which the multicast data is sourced.

```

1 regress@r1-sb-bier-RE0> show pim join extensive instance m_customer_A inet
2
3 Warning: License key missing; requires 'PIM' license
4
5 Instance: PIM.m_customer_A Family: INET
6 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
7
8 Group: 232.1.1.1
9   Source: 13.1.1.2
10  Flags: sparse
11  Upstream interface: et-0/0/18.10
12  Upstream neighbor: Direct
13  Upstream state: Local Source
14  Keepalive timeout:
15  Uptime: 20:32:50
16  Downstream neighbors:
17    Interface: Pseudo-MVPN
18      Uptime: 20:32:50 Time since last Join: 20:32:50
19    Number of downstream interfaces: 1
20    Number of downstream neighbors: 1
21
22 regress@r1-sb-bier-RE0>

```

1.1.5 MVPN Type-1 Route Advertisements

Through the type-1 routes, the peering routers PE or BFIR/BFER advertise their PMSI characteristics including VPN label, subdomain id, and bfr-id.

Let's analyze the routes advt. by the source PE for the VPN customer. As shown below, at the receiver PE, R4, it received the VPN information from the source PE, R1 with VPN label as 990001, Sub Domain 100, BFR-ID 1.

```

1 regress@r4-sb-bier-RE0> show route receive-protocol bgp 11.1.1.2 table bgp.mvpn.0
2
3 Warning: License key missing; requires 'BGP' license
4
5 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
6   Prefix Nexthop MED Lclpref AS path
7   1:11.1.1.1:1:11.1.1.1/240
8 * 11.1.1.1 100 I
9   1:11.1.1.5:1:11.1.1.5/240
10 * 11.1.1.5 100 I
11  1:11.1.1.6:1:11.1.1.6/240
12 * 11.1.1.6 100 I
13  1:11.1.1.7:1:11.1.1.7/240
14 * 11.1.1.7 100 I
15
16 regress@r4-sb-bier-RE0>
17
18 regress@r4-sb-bier-RE0> show route receive-protocol bgp 11.1.1.2 table bgp.mvpn.0 match-prefix
19   1:11.1.1.1:1:11.1.1.1/240 extensive
20
21 Warning: License key missing; requires 'BGP' license
22
23 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
24 * 1:11.1.1.1:1:11.1.1.1/240 (1 entry, 0 announced)
25   Import Accepted
26   Route Distinguisher: 11.1.1.1:1
27   Nexthop: 11.1.1.1
28   Localpref: 100
29   AS path: I (Originator)
30   Cluster list: 11.1.1.2
31   Originator ID: 11.1.1.1
32   Communities: target:64512:1

```

```

32      PMSI: Flags 0x0: Label 990001: Type BIER 11.1.1.1 100 1
33
34 regress@r4-sb-bier-RE0>
35
36 regress@r4-sb-bier-RE0> show route table bgp.mvpn.0 match-prefix 1:11.1.1.1:1:11.1.1.1/240
    extensive
37
38 bgp.mvpn.0: 15 destinations, 15 routes (15 active, 0 holddown, 0 hidden)
39 1:11.1.1.1:1:11.1.1.1/240 (1 entry, 0 announced)
40 *BGP Preference: 170/-101
41      PMSI: Flags 0x0: Label 990001: Type BIER 11.1.1.1 100 1
42      Next hop type: Indirect, Next hop index: 0
43      Address: 0x55cf71960a5c
44      Next-hop reference count: 12
45      Kernel Table Id: 0
46      Source: 11.1.1.2
47      Protocol next hop: 11.1.1.1
48      Indirect next hop: 0x2 no-forward INH Session ID: 0
49      Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
50      State: <Active Int Ext>
51      Local AS: 64512 Peer AS: 64512
52      Age: 2d 17:54:26 Metric2: 1
53      Validation State: unverified
54      Task: BGP_64512.11.1.1.2
55      AS path: I (Originator)
56      Cluster list: 11.1.1.2
57      Originator ID: 11.1.1.1
58      Communities: target:64512:1
59      Import Accepted
60      Localpref: 100
61      Router ID: 11.1.1.2
62      Secondary Tables: m_customer_A.mvpn.0
63      Thread: junos-main
64      Indirect next hops: 1
65          Protocol next hop: 11.1.1.1 Metric: 1 ResolvState: Resolved
66          Indirect next hop: 0x2 no-forward INH Session ID: 0
67          Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
68          Indirect path forwarding next hops: 1
69              Next hop type: Router
70              Next hop: 12.1.4.1 via et-0/0/9.0
71              Session Id: 3
72              Statistics ID Group: Kernel ID = 4123, Stats IDs = { 805306391 }
73              11.1.1.1/32 Originating RIB: inet.3
74              Metric: 1 Node path count: 1
75              Forwarding nexthops: 1
76                  Next hop type: Router
77                  Next hop: 12.1.4.1 via et-0/0/9.0
78                  Session Id: 3
79                  Statistics ID Group: Kernel ID = 4123, Stats IDs = { 805306391 }
80
81 regress@r4-sb-bier-RE0>

```

On receiving this information at the receiver PE, R4, it adds the label to the mpls.0. When it receives a packet with VPN label of 990001, it will send it to the tunnel ifl lsi.0 after popping the label.

```

1 regress@r4-sb-bier-RE0> show route label 990001
2
3 mpls.0: 18 destinations, 18 routes (18 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 990001 *[VPN/0] 21:28:55
7     > via lsi.0 (m_customer_A), Pop
8
9 regress@r4-sb-bier-RE0> show route label 990001 extensive
10
11 mpls.0: 18 destinations, 18 routes (18 active, 0 holddown, 0 hidden)
12 990001 (1 entry, 1 announced)
13 TSI:
14 KRT in-kernel 990001 /52 -> {Pop }
15 *VPN Preference: 0
16     Next hop type: Router, Next hop index: 9050
17     Address: 0x55b7f336e25c
18     Next-hop reference count: 2, Next-hop session id: 1
19     Kernel Table Id: 0
20     Next hop: via lsi.0 (m_customer_A), selected
21     Label operation: Pop
22     Load balance label: None;
23     Label element ptr: 0x55b7f3257f60
24     Label parent element ptr: (nil)

```

```

25      Label element references: 1
26      Label element child references: 0
27      Label element lsp id: 0
28      Session Id: 1
29      State: <Active Int Ext LsiL3>
30      Age: 21:28:58
31      Validation State: unverified
32      Task: RT
33      Announcement bits (1): 1-KRT
34      AS path: I
35      Thread: junos-main
36
37 regress@r4-sb-bier-RE0>

```

Let's analyze the routes advertised by the receiver PE, R4, to other PEs (to the source PE, R1)

```

1 regress@r4-sb-bier-RE0> show route advertising-protocol bgp 11.1.1.2 table bgp.mvpn.0
2
3 Warning: License key missing; requires 'BGP' license
4
5 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
6   Prefix Nexthop MED LcIpref AS path
7   1:11.1.1.4:1:11.1.1.4/240
8   * Self 100 I
9   7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
10  * Self 0 100 I
11
12 regress@r4-sb-bier-RE0>
13
14 regress@r4-sb-bier-RE0> show route advertising-protocol bgp 11.1.1.2 table bgp.mvpn.0 match-
    prefix 1:11.1.1.4:1:11.1.1.4/240
15
16 Warning: License key missing; requires 'BGP' license
17
18 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
19   Prefix Nexthop MED LcIpref AS path
20   1:11.1.1.4:1:11.1.1.4/240
21   * Self 100 I
22
23 regress@r4-sb-bier-RE0>

```

On the Source PE's bgp.mvpn.0 table, we could see the type-1 routes advertised by the receiver PE, PE4 as follows:

```

1 regress@r1-sb-bier-RE0> show route table bgp.mvpn.0 match-prefix 1:11.1.1.4:1:11.1.1.4/240
2
3 bgp.mvpn.0: 6 destinations, 7 routes (6 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 1:11.1.1.4:1:11.1.1.4/240
7      *[BGP/170] 21:21:51, localpref 100, from 11.1.1.2
8      AS path: I, validation-state: unverified
9      > to 12.1.1.2 via et-0/0/9.0, Push 22
10
11 regress@r1-sb-bier-RE0>
12
13 regress@r1-sb-bier-RE0> show route table bgp.mvpn.0 match-prefix 1:11.1.1.4:1:11.1.1.4/240
    extensive
14
15 bgp.mvpn.0: 6 destinations, 7 routes (6 active, 0 holddown, 0 hidden)
16 1:11.1.1.4:1:11.1.1.4/240 (1 entry, 0 announced)
17   *BGP Preference: 170/-101
18     PMSI: Flags 0x0: Label 990001: Type BIER 11.1.1.4 100 4
19     Next hop type: Indirect, Next hop index: 0
20     Address: 0x55885715a69c
21     Next-hop reference count: 2
22     Kernel Table Id: 0
23     Source: 11.1.1.2
24     Protocol next hop: 11.1.1.4
25     Indirect next hop: 0x2 no-forward INH Session ID: 0
26     Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
27     State: <Active Int Ext>
28     Local AS: 64512 Peer AS: 64512
29     Age: 21:22:44 Metric2: 1
30     Validation State: unverified
31     Task: BGP_64512.11.1.1.2
32     AS path: I (Originator)
33     Cluster list: 11.1.1.2

```

```

34      Originator ID: 11.1.1.4
35      Communities: target:64512:1
36      Import Accepted
37      Localpref: 100
38      Router ID: 11.1.1.2
39      Secondary Tables: m_customer_A.mvpn.0
40      Thread: junos-main
41      Indirect next hops: 1
42          Protocol next hop: 11.1.1.4 Metric: 1 ResolvState: Resolved
43          Indirect next hop: 0x2 no-forward INH Session ID: 0
44          Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
45          Indirect path forwarding next hops: 1
46              Next hop type: Router
47              Next hop: 12.1.1.2 via et-0/0/9.0
48              Session Id: 2
49              Statistics ID Group: Kernel ID = 4106, Stats IDs = { 805306379 }
50              11.1.1.4/32 Originating RIB: inet.3
51              Metric: 1 Node path count: 1
52              Forwarding nexthops: 1
53                  Next hop type: Router
54                  Next hop: 12.1.1.2 via et-0/0/9.0
55                  Session Id: 2
56                  Statistics ID Group: Kernel ID = 4106, Stats IDs = { 805306379 }
57
58 regress@r1-sb-bier-RE0>

```

The type-1 routes protocol next-hop is BFER R4, reachable as follows:

```

1 regress@r1-sb-bier-RE0> show route 11.1.1.4 table inet.3
2
3 inet.3: 15 destinations, 15 routes (15 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 11.1.1.4/32 *[LDP/9] 21:23:57, metric 1
7     > to 12.1.1.2 via et-0/0/9.0, Push 22
8
9 regress@r1-sb-bier-RE0>

```

This implies that, for the BGP MVPN peering to work, we need label exchange protocols like LDP, RSVP, and SR in the core. In our case, we enabled LDP, which provides the labels to inet.3, which BGP uses for resolution.

Note: For the BIER domain to work, the labels are advertised by the IGP underlay (ISIS). The overlay (MVPN) depends on the labeling protocol LDP, RSVP, or SR.

1.1.6 Type-7 Route Advertisement

The receiving PE router advertises the type-7 to source PE:

```

1 regress@r4-sb-bier-RE0> show route advertising-protocol bgp 11.1.1.2 table bgp.mvpn.0 match-
2     prefix 7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
3 Warning: License key missing; requires 'BGP' license
4
5 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
6   Prefix Nexthop MED Lclpref AS path
7   7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
8   * Self 0 100 I
9
10 regress@r4-sb-bier-RE0>
11
12 regress@r4-sb-bier-RE0> ...table bgp.mvpn.0 match-prefix
13     7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240 extensive
14 Warning: License key missing; requires 'BGP' license
15
16 bgp.mvpn.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
17 * 7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240 (1 entry, 1 announced)
18   BGP group bier_mvpn type Internal
19   Route Distinguisher: 11.1.1.1:1
20   Nexthop: Self
21   Flags: Nexthop Change
22   MED: 0
23   Localpref: 100

```

```

24 AS path: [64512] I
25 Communities: target:11.1.1.1:51
26
27 regress@r4-sb-bier-RE0>

```

Let's check the routes in the source PE's bgp.mvpn.0 table

```

1 regress@r1-sb-bier-RE0> show route table bgp.mvpn.0 match-prefix
  7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
2
3 bgp.mvpn.0: 6 destinations, 7 routes (6 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
7      *[PIM/105] 21:02:20, metric 0
8      Multicast (IPv4) Composite
9      [BGP/170] 21:02:20, MED 0, localpref 100, from 11.1.1.2
10     AS path: I, validation-state: unverified
11     > to 12.1.1.2 via et-0/0/9.0
12
13 regress@r1-sb-bier-RE0>
14
15 regress@r1-sb-bier-RE0> show route table bgp.mvpn.0 match-prefix
  7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240 extensive
16
17 bgp.mvpn.0: 6 destinations, 7 routes (6 active, 0 holddown, 0 hidden)
18 7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240 (2 entries, 0 announced)
19      *PIM Preference: 105
20      Next hop type: Multicast (IPv4) Composite, Next hop index: 0
21      Address: 0x7f076adcc18c
22      Next-hop reference count: 4
23      Kernel Table Id: 0
24      State: <Secondary Active Int Ext>
25      Age: 21:02:56 Metric: 0
26      Validation State: unverified
27      Task: PIM.m_customer_A
28      AS path: I
29      Communities: no-advertise target:11.1.1.1:51
30      Primary Routing Table: m_customer_A.mvpn.0
31      Thread: junos-main
32      BGP Preference: 170/-101
33      Next hop type: Indirect, Next hop index: 0
34      Address: 0x55885715d4fc
35      Next-hop reference count: 2
36      Kernel Table Id: 0
37      Source: 11.1.1.2
38      Protocol next hop: 11.1.1.2
39      Indirect next hop: 0x2 no-forward INH Session ID: 0
40      Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
41      State: <Int Ext>
42      Inactive reason: Route Preference
43      Local AS: 64512 Peer AS: 64512
44      Age: 21:02:56 Metric: 0 Metric2: 1
45      Validation State: unverified
46      Task: BGP_64512.11.1.1.2
47      AS path: I (Originator)
48      Cluster list: 11.1.1.2
49      Originator ID: 11.1.1.2
50      Communities: target:11.1.1.1:51
51      Import Accepted
52      Localpref: 100
53      Router ID: 11.1.1.2
54      Secondary Tables: m_customer_A.mvpn.0
55      Thread: junos-main
56      Indirect next hops: 1
57          Protocol next hop: 11.1.1.2 Metric: 1 ResolvState: Resolved
58          Indirect next hop: 0x2 no-forward INH Session ID: 0
59          Indirect next hop: INH non-key opaque: (nil) INH key opaque: (nil)
60          Indirect path forwarding next hops: 1
61              Next hop type: Router
62              Next hop: 12.1.1.2 via et-0/0/9.0
63              Session Id: 0
64              Statistics ID Group: Kernel ID = 0, Stats IDs = { 805306375 }
65              11.1.1.2/32 Originating RIB: inet.3
66              Metric: 1 Node path count: 1
67              Forwarding nexthops: 1
68                  Next hop type: Router
69                  Next hop: 12.1.1.2 via et-0/0/9.0
70                  Session Id: 0
71                  Statistics ID Group: Kernel ID = 0, Stats IDs = { 805306375 }

```

```

72
73 regress@r1-sb-bier-RE0>

```

Let's check the routes in VPN table:

```

1 regress@r1-sb-bier-RE0> show route table m_customer_A.mvpn.0 match-prefix
  7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
2
3 m_customer_A.mvpn.0: 6 destinations, 7 routes (6 active, 1 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 7:11.1.1.1:1:64512:32:13.1.1.2:32:232.1.1.1/240
7      *[PIM/105] 21:04:33, metric 0
8          Multicast (IPv4) Composite
9          [BGP/170] 21:04:33, MED 0, localpref 100, from 11.1.1.2
10         AS path: I, validation-state: unverified
11         > to 12.1.1.2 via et-0/0/9.0
12
13 regress@r1-sb-bier-RE0>
14
15 regress@r1-sb-bier-RE0> show mvpn instance inet
16
17 MVPN instance:
18 Legend for neighbor state (St)
19 A- Preferred upstream neighbor for inter-AS
20
21 Legend for provider tunnel
22 S- Selective provider tunnel
23 F- Flood NH forwarding NH
24 M- Multicast Composite NH
25 C- Cloned NH
26
27 Legend for c-multicast routes properties (St)
28 DS -- derived from (*, c-g) RM -- remote VPN route
29 I -- Inactive
30 Family : INET
31
32 Instance : m_customer_A
33 MVPN Mode : SPT-ONLY
34 Sender-Based RPF: Disabled. Reason: Not enabled by configuration.
35 Hot Root Standby: Disabled. Reason: Not enabled by configuration.
36 Provider tunnel: I-P-tnl:BIER: subdomain-id 100 bfr-id 1 label 990001
37 Neighbor Inclusive Provider Tunnel Label-In St Segment
38 11.1.1.4 BIER: subdomain-id 100 bfr-id 4 bier-prefix 11.1.1.4 990001
39 11.1.1.5 BIER: subdomain-id 100 bfr-id 5 bier-prefix 11.1.1.5 990001
40 11.1.1.6 BIER: subdomain-id 100 bfr-id 6 bier-prefix 11.1.1.6 990001
41 11.1.1.7 BIER: subdomain-id 100 bfr-id 7 bier-prefix 11.1.1.7 990001
42
43 C-mcast IPv4 (S:G) Provider Tunnel Label-In St FwdNh Segment
44 13.1.1.2/32:232.1.1.1/32 BIER: subdomain-id 100 bfr-id 1 label 990001 RM M-8063
45
46 regress@r1-sb-bier-RE0>

```

As shown below at the BFIR, the multicast packet is encapsulated with a VPN label of 990001, the BIER BitString containing BFR-IDs of R4, R5, R6, and R7, and the BIER label of 16 which corresponds to the {ISIS Topology: Default, BIER Sub-Domain:100, Set:0}.

```

1 regress@r1-sb-bier-RE0> show multicast route instance m_customer_A inet
2 Instance: m_customer_A Family: INET
3
4 Group: 232.1.1.1
5   Source: 13.1.1.2/32
6   Upstream interface: et-0/0/18.10
7   Downstream interface list:
8     Push 990001, BS:0:0:0:0:0:0:0:0:00000078, label 16
9
10 regress@r1-sb-bier-RE0> show multicast route instance m_customer_A inet extensive
11 Instance: m_customer_A Family: INET
12
13 Group: 232.1.1.1
14   Source: 13.1.1.2/32
15   Upstream interface: et-0/0/18.10
16   Downstream interface list:
17     Push 990001, BS:0:0:0:0:0:0:0:0:00000078, label 16
18   Number of outgoing interfaces: 1
19   Session description: Source specific multicast
20   Statistics: 51 kbps, 100 pps, 475555 packets

```

```

21 Next-hop ID: 6009
22 Upstream protocol: MVPN
23 Route state: Active
24 Forwarding state: Forwarding
25 Cache lifetime/timeout: forever
26 Wrong incoming interface notifications: 0
27 Uptime: 21:07:19
28 Sensor ID: 0xf000000a
29
30 regress@r1-sb-bier-RE0>
31
32 regress@r1-sb-bier-RE0> show route table mpls.0 protocol bier
33
34 mpls.0: 18 destinations, 18 routes (18 active, 0 holddown, 0 hidden)
35 + = Active Route, - = Last Active, * = Both
36
37 16 *[BIER/70] 21:35:39
38     Multicast (IPv4) Composite
39 17 *[BIER/70] 22:01:00
40     Discard
41 18 *[BIER/70] 22:01:00
42     Discard
43 19 *[BIER/70] 22:01:00
44     Discard
45
46 regress@r1-sb-bier-RE0>

```

1.2 BIER for IPv6 Multicast

BIER carries the IPv6 multicast data over the Service Provider Core Network. In the below topology IPv6 Listeners sends MLD join from R1 and R7. The IPv6 multicast data stream is behind R4.

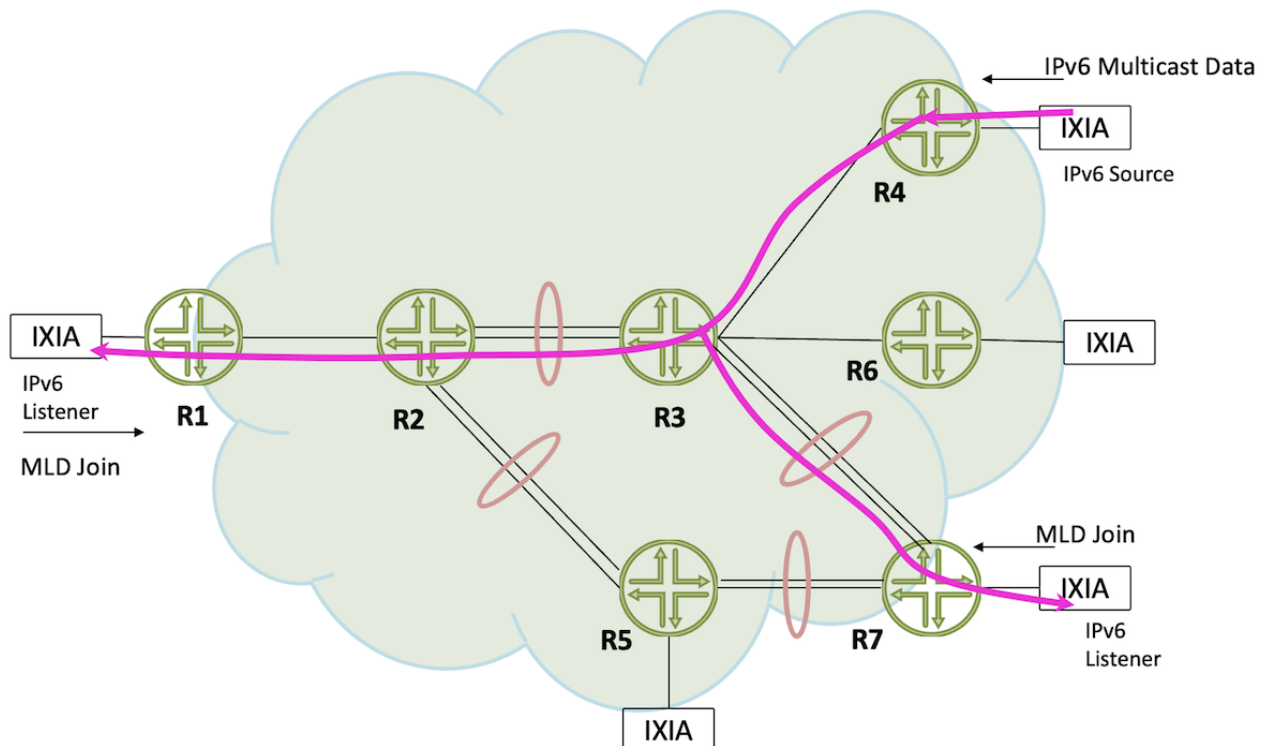


Figure 2: BIER IPv6 Multicast Flow

As shown below the IPv6 listener behind R7 sends MLD join for the SSM group ff3a::1

```

1 regress@r7-sb-bier-RE0> show mld group
2 Interface: et-0/0/11.30, Groups: 3
3 Group: ff02::1:ff00:1

```

```

 4      Source: ::
 5      Last reported by: fe80::21c:1ff:fe00:1
 6      Timeout: 142 Type: Dynamic
 7      Group: ff02::1:ff00:2
 8      Source: ::
 9      Last reported by: fe80::21c:1ff:fe00:1
10      Timeout: 142 Type: Dynamic
11      Group: ff3a::1
12      Group mode: Include
13      Source: 2002:13:3:2::2
14      Last reported by: fe80::21c:1ff:fe00:1
15      Timeout: 140 Type: Dynamic
16 Interface: local, Groups: 4
17   Group: ff02::
18   Source: ::
19   Last reported by: Local
20   Timeout: 0 Type: Dynamic
21   Group: ff02::2
22   Source: ::
23   Last reported by: Local
24   Timeout: 0 Type: Dynamic
25   Group: ff02::d
26   Source: ::
27   Last reported by: Local
28   Timeout: 0 Type: Dynamic
29   Group: ff02::16
30   Source: ::
31   Last reported by: Local
32   Timeout: 0 Type: Dynamic
33
34 regress@r7-sb-bier-RE0>

```

The BFER R7, advertises a Type-7 route corresponding to the SSM group of interest.

```

 1 regress@r7-sb-bier-RE0> show route advertising-protocol bgp 11.1.1.2 table bgp.mvpn-inet6.0
 2
 3 Warning: License key missing; requires 'BGP' license
 4
 5 bgp.mvpn-inet6.0: 10 destinations, 10 routes (10 active, 0 holddown, 0 hidden)
 6   Prefix Nexthop MED Lclpref AS path
 7   1:11.1.1.7:1:11.1.1.7/432
 8   * Self 100 I
 9   1:11.1.1.7:2:11.1.1.7/432
10   * Self 100 I
11   1:11.1.1.7:3:11.1.1.7/432
12   * Self 100 I
13   7:11.1.1.4:3:64512:128:2002:13:3:2::2:128:ff3a::1/432
14   * Self 0 100 I
15
16 regress@r7-sb-bier-RE0>

```

The extensive output shows the next-hop type as IPv6 Composite Multicast

```

 1 regress@r7-sb-bier-RE0> show route table bgp.mvpn-inet6.0 match-prefix
 2   7:11.1.1.4:3:64512:128:2002:13:3:2::2:128:ff3a::1/432 extensive
 3
 4 bgp.mvpn-inet6.0: 10 destinations, 10 routes (10 active, 0 holddown, 0 hidden)
 5 7:11.1.1.4:3:64512:128:2002:13:3:2::2:128:ff3a::1/432 (1 entry, 1 announced)
 6 TSI:
 7 Page 0 idx 0, (group bier_mvpn type Internal) Type 1 val 0x563f39fc6760 (adv_entry)
 8   Advertised metrics:
 9     Flags: Nexthop Change
10     Nexthop: Self
11     MED: 0
12     Localpref: 100
13     AS path: [64512] I
14     Communities: target:11.1.1.4:54
15     Advertise: 00000001
16 Path 7:11.1.1.4:3:64512:128:2002:13:3:2::2:128:ff3a::1
17 Vector len 4. Val: 0
18   *PIM Preference: 105
19     Next hop type: Multicast (IPv6) Composite, Next hop index: 0
20     Address: 0x563f2e81ad84
21     Next-hop reference count: 2
22     Kernel Table Id: 0
23     State: <Secondary Active Int Ext>
24     Age: 36 Metric: 0
25     Validation State: unverified
26     Task: PIM.m_customer_C

```

```

26      Announcement bits (1): 0-BGP_RT_Background
27      AS path: I
28      Communities: target:11.1.1.4:54
29      Primary Routing Table: m_customer_C.mvpn-inet6.0
30      Thread: junos-main

```

The multicast route at the BFIR R4 captures the downstream BIER tunnel

```

1 regress@r4-sb-bier-RE0> show multicast route instance m_customer_C
2 Instance: m_customer_C Family: INET
3
4 Instance: m_customer_C Family: INET6
5
6 Group: ff3a::1
7   Source: 2002:13:3:2::2/128
8   Upstream interface: et-0/0/18.30
9   Downstream interface list:
10     Push 990003, BS:0:0:0:0:0:0:00000041, label 16
11
12 regress@r4-sb-bier-RE0> show multicast route instance m_customer_C extensive
13 Instance: m_customer_C Family: INET
14
15 Instance: m_customer_C Family: INET6
16
17 Group: ff3a::1
18   Source: 2002:13:3:2::2/128
19   Upstream interface: et-0/0/18.30
20   Downstream interface list:
21     Push 990003, BS:0:0:0:0:0:0:00000041, label 16
22   Number of outgoing interfaces: 1
23   Session description: Unknown
24   Statistics: 51 kbps, 100 pps, 9855 packets
25   Next-hop ID: 6035
26   Upstream protocol: MVPN
27   Route state: Active
28   Forwarding state: Forwarding
29   Cache lifetime/timeout: forever
30   Wrong incoming interface notifications: 0
31   Uptime: 00:01:38
32   Sensor ID: 0xf0000028
33
34 regress@r4-sb-bier-RE0>

```

1.3 BIER Multicast Packet Flow

BIER includes the functionality provided by three layers:

- The BIER underlay, in our case ISIS, gives the necessary information to create the BIER Routing Tables (BIRT) and BIER Forwarding Tables (BIFT) with information FBM (Forwarding Bit Mask) and BIER labels.
- The BIER layer, states the encapsulation to pick and the various BSL (Bit string lengths) and Sets.
- The BIER Overlay, in our case MVPN, the MVPN Overlay creates the bitstring to be appended to the multicast packet to be sent to the core from the IBGP MVPN peering with other PE routers of the core and builds either inclusive or selective tunnels. It also appends the correct VPN label for the destination VPN customers.

When the entire BFER falls into the same set, the multicast data replication is as follows. Here multicast source is behind R1 and multicast listeners are at R4, R5, R6 and R7

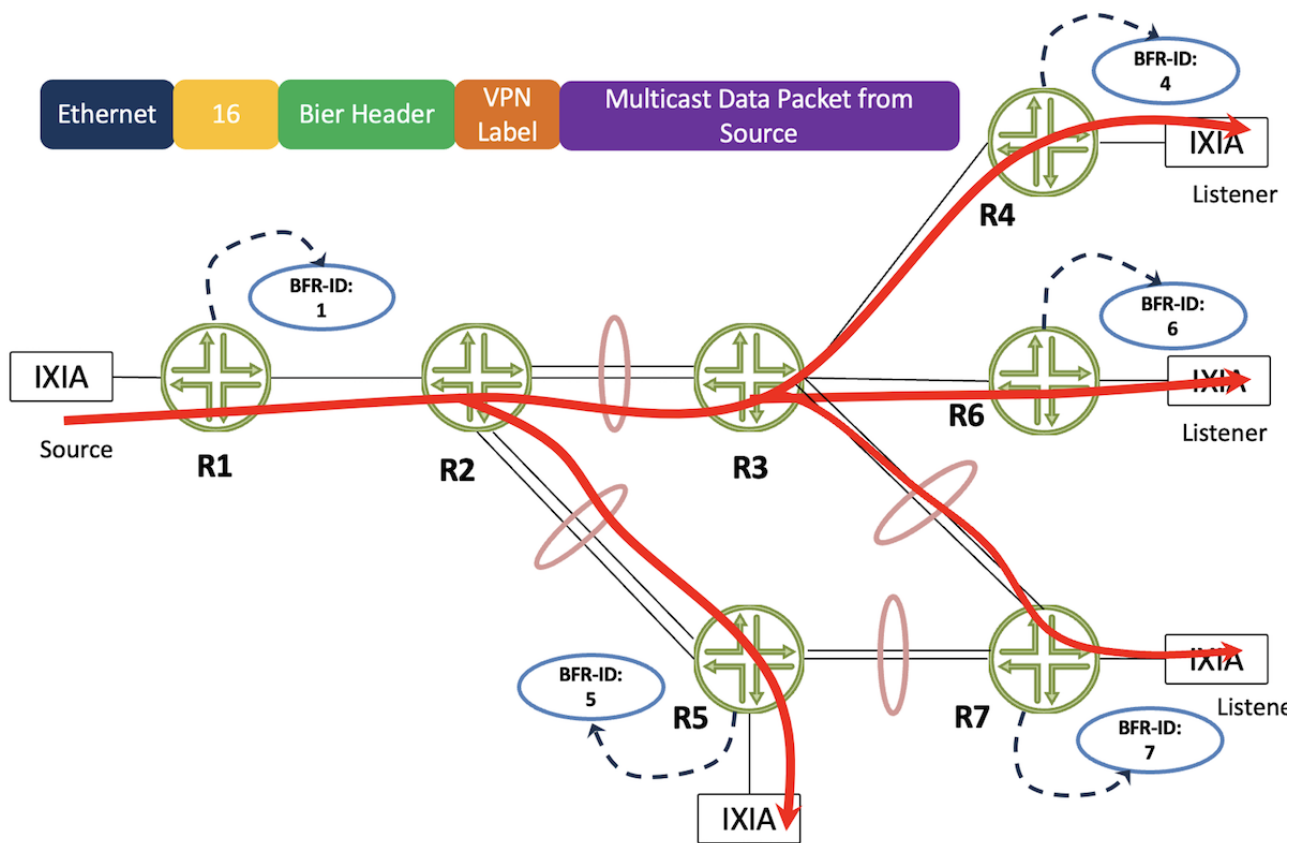


Figure 3: BIER Multicast Flow

Let's get the traffic stats at the BFR R1. It shows 100pps packets sent toward R2 for the FBM

```
1 regress@r1-sb-bier-RE0> show bier neighbor
2 BIER info:
3 Subdomain ID: 100 BSL: 256
4 BIER Neighbor Label Nexthop Packets(pps) Bytes
5 11.1.1.2 16 12.1.1.2 9340355( 100) 5025110990
6 FBM:0:0:0:0:0:0:0:0:00000078
7
8 regress@r1-sb-bier-RE0>
```

At BFR R2, we have three neighbors. Two of them are toward the listeners

```
1 regress@r2-sb-bier-RE0> show bier neighbor
2 BIER info:
3 Subdomain ID: 100 BSL: 256
4 BIER Neighbor Label Nexthop Packets(pps) Bytes
5 11.1.1.1 16 12.1.1.1 0( 0) 0
6 FBM:0:0:0:0:0:0:0:0:00000001
7 11.1.1.3 16 12.1.2.2 597729( 100) 321578202
8 FBM:0:0:0:0:0:0:0:0:00000068
9 11.1.1.5 16 12.1.3.2 597107( 100) 321243566
10 FBM:0:0:0:0:0:0:0:0:00000050
11
12 regress@r2-sb-bier-RE0>
```

At BFR R3, we have three BFER neighbors

```
1 regress@r3-sb-bier-RE0> show bier neighbor
2 BIER info:
3 Subdomain ID: 100 BSL: 256
4 BIER Neighbor Label Nexthop Packets(pps) Bytes
5 11.1.1.4 16 12.1.4.2 15214713( 100) 8185515594
6 FBM:0:0:0:0:0:0:0:0:00000008
7 11.1.1.6 16 12.1.7.2 15219117( 100) 8187884946
8 FBM:0:0:0:0:0:0:0:0:00000020
9 11.1.1.7 16 12.1.6.2 14316592( 100) 7702326496
```

```

10 FBM:0:0:0:0:0:0:0:0:00000050
11 11.1.1.2 26 12.1.2.1 0( 0) 0
12 FBM:0:0:0:0:0:0:0:0:00000011
13
14 regress@r3-sb-bier-RE0>

```

The BIER multicast packet structure within the BIER domain is as follows:



Figure 4: Multicast Data with BIER header

When the BFER are in different sets, one copy of the multicast data is created for each of the sets as shown below. **

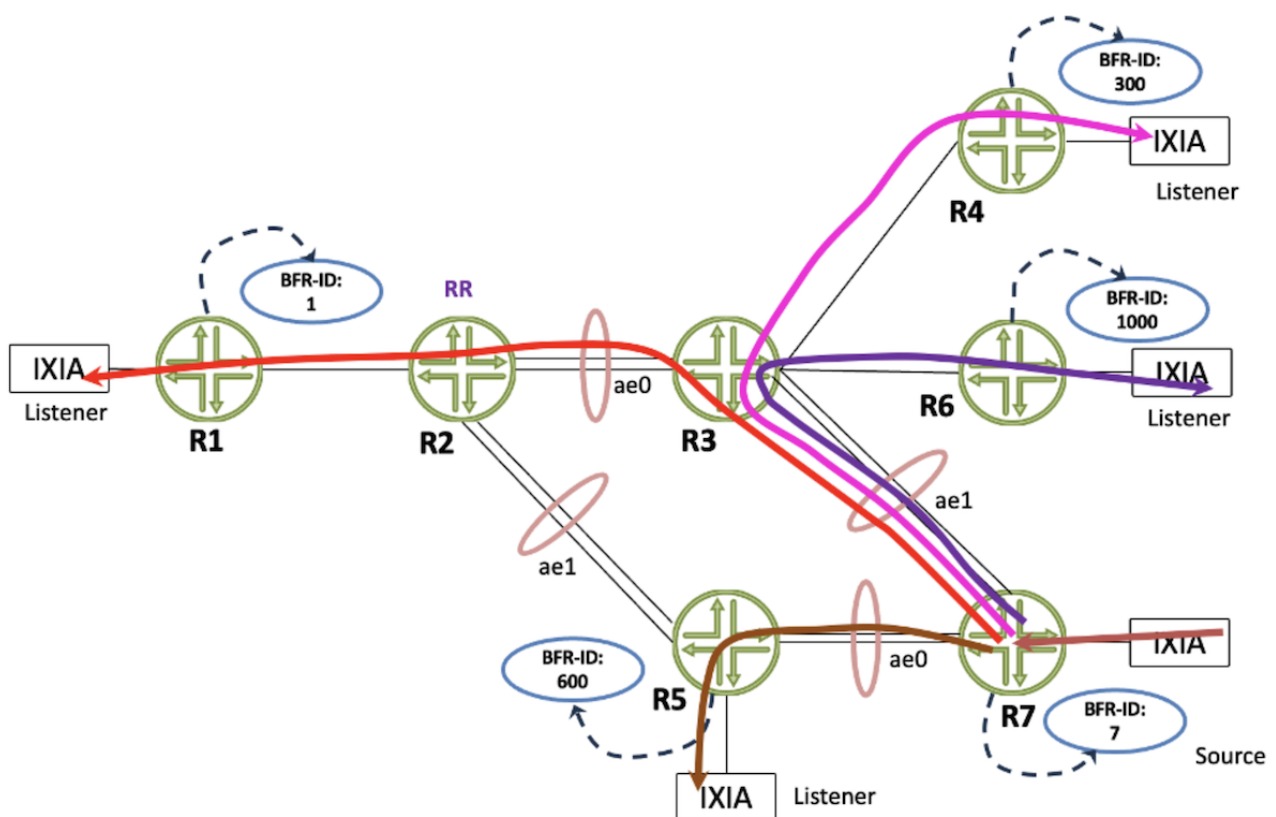


Figure 5: BIER Multicast Flow with different sets

The multicast data packet structure for each of the sets is as follows:

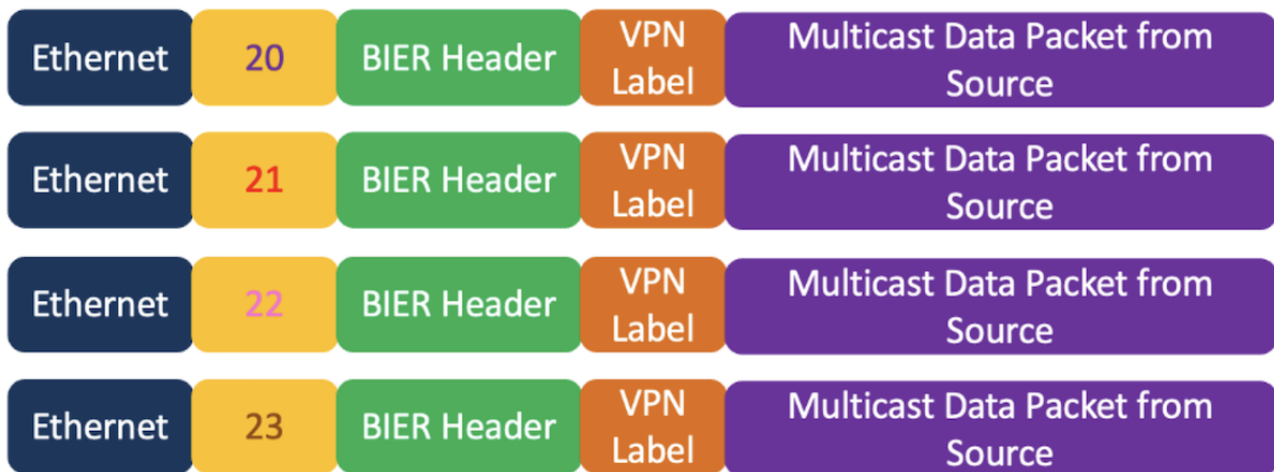


Figure 6: BIER Multicast Packet Headers for different sets

The following table captures the multicast replication for the BFER in different subdomains

```

1 regress@r7-sb-bier-RE0> show bier neighbor subdomain-id 200
2 BIER info:
3 Subdomain ID: 200 BSL: 256
4 BIER Neighbor Label Nexthop Packets(pps) Bytes
5 11.1.1.3 20 12.1.6.1 468550( 108) 252079900
6 FBM:0:0:0:0:0:0:0:0:00000001
7 11.1.1.5 20 12.1.8.1 0( 0) 0
8 FBM:0:0:0:0:0:0:0:0:00000001
9 11.1.1.3 21 12.1.6.1 624163( 108) 335799694
10 FBM:0:0:0:0:0:0:0:0:00000800:0
11 11.1.1.5 22 12.1.8.1 597120( 108) 321250560
12 FBM:0:0:0:0:0:0:0:0:00800000:0:0
13 11.1.1.3 23 12.1.6.1 624165( 108) 335800770
14 FBM:00000080:0:0:0:0:0:0:0
15
16 regress@r7-sb-bier-RE0>

```

In the following example, the multicast source is behind R7 and listeners are behind R1, R4, R5, and R6 in different sets, we have four downstream tunnels.

```

1 regress@r7-sb-bier-RE0> show multicast route instance m_customer_B fa inet
2 Instance: m_customer_B Family: INET
3
4 regress@r7-sb-bier-RE0> show multicast route instance m_customer_B
5 Instance: m_customer_B Family: INET
6
7 Group: 232.2.2.2
8 Source: 13.2.5.2/32
9 Upstream interface: et-0/0/11.20
10 Downstream interface list:
11 Push 990002, BS:0:0:0:0:0:0:0:0:00800000:0:0, label 22 Push 990002, BS:00000080:0:0:0:0:0:0:0,
12 label 23
13 Push 990002, BS:0:0:0:0:0:0:0:0:00000001, label 20 Push 990002, BS:0:0:0:0:0:0:0:0:00000800:0,
14 label 21

```

The multicast route info from BFER perspective is as follows, per multicast VPN, there will be one lsi interface created and that will act as the upstream interface towards the source of the multicast stream.

```

1 regress@r1-sb-bier-RE0> show multicast route instance m_customer_B
2 Instance: m_customer_B Family: INET
3
4 Group: 232.2.2.2
5 Source: 13.2.5.2/32
6 Upstream interface: lsi.1
7 Downstream interface list:
8 et-0/0/18.20

```

1.4 Useful links

RFC/Drafts

- <https://datatracker.ietf.org/doc/rfc8279/>
- <https://datatracker.ietf.org/doc/rfc8296/>
- <https://datatracker.ietf.org/doc/rfc8401/>
- <https://datatracker.ietf.org/doc/rfc8556/>
- <https://datatracker.ietf.org/doc/rfc9573/>
- <https://www.rfc-editor.org/rfc/rfc7902.html>

PTX10002-36QDD:

- <https://juniper.github.io/techposts/introducing-ptx10002-36qdd/article>
- <https://juniper.github.io/techposts/express-5-overview/article>
- <https://www.juniper.net/us/en/products/routers/ptx-series/ptx10002-36qdd-packet-transport-router.html>

BIER TechPost Articles

- <https://juniper.github.io/techposts/bier-mvpn-in-ptx-express-5/article>
- <https://juniper.github.io/techposts/eantc-bier-interop-testing-on-ptx10002-36qdd/article>

EANTC 2024 InterOP Report

- <https://eantc.de/wp-content/uploads/2023/12/EANTC-MPLSSDNInterop2024-TestReport-v1.3.pdf>

1.5 Glossary

- BIER Bit Index Explicit Replication
- BFR Bit Forwarding Router
- BFR-NBR BFR Neighbor
- BFR-Prefix BFR Prefix IP address
- BFER Bit Forwarding Egress Router
- BFIR Bit Forwarding Ingress Router
- BIRT Bit Index Routing Table
- BIFT Bit Index Forwarding Table
- BML Bit Mask Length
- BSL Bit String Length
- BFR-Id BFR Identifier
- BGP Border Gateway Protocol
- DCB Domainwide Common Block
- DLU Destination Lookup
- EANTC European Advanced Networking Test Center
- EPP Egress Packet Processor
- FBM Forwarding Bit Mask
- IGMP Internet Group Management Protocol
- IGP Interior Gateway Protocol

- IGP Ingress Parser
- ISIS Intermediate System Intermediate System
- LDP Label Distribution Protocol
- LSA Link State Advertisement
- MCE Multicast Engine
- m-LDP Multicast LDP
- MPLS Multi Protocol Label Switching
- MVPN Multicast VPN
- MWC MPLS World Congress
- OSPF Open Shortest Path First
- PIM Protocol Independent Multicast
- PMSI Provider Multicast Service Interface
- RFC Request For Comments
- RSVP-TE Resource Reservation Protocol – Traffic Engineering
- VRF Virtual Routing and Forwarding
- VPN Virtual Private Network

1.6 Acknowledgments

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